

Submission A solution to Problem 4

Partial proof: the first alternative and the remaining hard estimate

Problem statement and interpretation

I write the small universal constant in the problem as κ , reserving k for k -dilation. Rectangles are oriented Euclidean products and degree 1 means relative degree 1 on $H_4(R, \partial R; \mathbb{Z}) \rightarrow H_4(S, \partial S; \mathbb{Z})$.

The current proof is complete through the first alternative and through a conditional reduction of the hard case. It is not a complete proof of the stated dichotomy, because the averaged tightening estimate (EAT) below remains unproved. The arguments are written for smooth transverse maps; piecewise smooth maps may be treated by the same current-slicing argument, or by smoothing on strata with an arbitrarily small C^1 -loss and then absorbing that loss into the universal constants.

Standard inputs

All constants are positive and universal. If $\text{Dil}_2(f) \leq 1$, then $\text{Dil}_3(f), \text{Dil}_4(f) \leq 1$: for singular values $\sigma_1 \geq \dots \geq \sigma_4$, the inequality $\sigma_1 \sigma_2 \leq 1$ implies $\sigma_2 \leq 1$ and $\sigma_1 \dots \sigma_j \leq \sigma_1 \sigma_2$ for $j \geq 2$.

We use Guth's map-of-pairs estimate [1, Estimate 1]. For $n = 4, k = 2, j = 1, l = 4$ it gives

$$R_1^2 \text{Vol}(R) \geq c_0 S_1^3 S_2 S_3 S_4. \quad (1)$$

We also use the small-volume relative isoperimetric estimate in rectangles [1, Theorem 3]: if Z is a relative 2-cycle in R with $A = \text{Mass } Z \leq c_* R_1 R_2$, then

$$\text{Fill}_R(Z) \leq C_* \max\{A^{3/2}, A^2/R_1\}.$$

Consequently

$$\text{Fill}_R(Z) \geq L \implies \text{Mass } Z \geq c \min\{R_1 R_2, L^{2/3}, (R_1 L)^{1/2}\}, \quad (2)$$

and, after decreasing c_* ,

$$\text{Mass } Z < c_* R_1 R_2 \implies \text{Fill}_R(Z) \leq C R_1 R_2^2. \quad (3)$$

The first alternative

Let Q be the concentric subrectangle of the (x_3, x_4) -face of S with side lengths $S_3/2, S_4/2$, and set

$$P_y = [0, S_1] \times [0, S_2] \times \{y\}, \quad y \in Q.$$

Lemma 1 (central target plane). *Every relative 3-chain in S with boundary P_y , $y \in Q$, has mass at least $cS_1S_2S_3$.*

Proof. Choose a 1-Lipschitz function ψ on the (x_3, x_4) -rectangle which vanishes on its boundary and satisfies $\psi \geq cS_3$ on Q . Put $\beta = \psi dx_1 \wedge dx_2$. Then $d\beta$ has comass at most 1, and Stokes' theorem gives

$$\text{Mass } Y \geq \left| \int_Y d\beta \right| = \left| \int_{P_y} \beta \right| \geq cS_1S_2S_3.$$

The relative boundary terms vanish because $dx_1 \wedge dx_2 = 0$ on the x_1, x_2 -faces and $\psi = 0$ on the x_3, x_4 -faces. $\square$

Proposition 1. *There is $c_2 > 0$ such that, under the hypotheses of the problem, if*

$$R_1R_2^2 \leq c_2S_1S_2S_3,$$

then $R_3R_4 \geq c_2S_3S_4$.

Proof. Let $T = [[R]]$ be the fundamental relative current and put $F = (f_3, f_4)$. Since f has relative degree 1, $f_{\#}T = [[S]]$ as a relative 4-current. For a.e. $y \in Q$, slicing naturality gives

$$f_{\#}\langle T, F, y \rangle = \langle [[S]], (x_3, x_4), y \rangle = \pm P_y.$$

Write $Z_y = \langle T, F, y \rangle$. Any relative filling of Z_y in R pushes forward, using $\text{Dil}_3(f) \leq 1$, to a filling of P_y in S . Lemma 1 therefore gives

$$\text{Fill}_R(Z_y) \geq cS_1S_2S_3.$$

If c_2 is sufficiently small, (3) forces $\text{Mass } Z_y \geq c_*R_1R_2$. Since $J_2F \leq \text{Dil}_2(f) \leq 1$, the coarea formula yields

$$\text{Vol}(R) \geq \int_Q \text{Mass } Z_y dy \geq cR_1R_2S_3S_4.$$

Cancelling R_1R_2 proves the proposition. $\square$

The unconditional partial hard-case bound

Assume $R_1 = \alpha S_1 \leq S_1$ and that the first conclusion fails with a small constant. Proposition 1 gives

$$R_1R_2^2 \geq cS_1S_2S_3. \tag{4}$$

For the slices above, (2) with $L = cS_1S_2S_3$ gives

$$\text{Mass } Z_y \geq c \min\{R_1R_2, (S_1S_2S_3)^{2/3}, (R_1S_1S_2S_3)^{1/2}\}.$$

Using (4) and $R_1 = \alpha S_1$, every term in the minimum is at least $c\alpha^{1/2}S_1(S_2S_3)^{1/2}$. Hence

$$\text{Vol}(R) \geq c\alpha^{1/2}S_1S_2^{1/2}S_3^{3/2}S_4. \quad (5)$$

On the other hand, (1) gives

$$\text{Vol}(R) \geq c\alpha^{-2}S_1S_2S_3S_4. \quad (6)$$

Optimizing between (5) and (6) gives

$$\text{Vol}(R) \geq cS_1S_2^{3/5}S_3^{7/5}S_4.$$

Thus the remaining gap is exactly the high-aspect factor $(S_3/S_2)^{1/10}$.

A sufficient averaged hard-case estimate

Let $G = (f_1, f_2)$, keep $F = (f_3, f_4)$, and define

$$E_y = \int_{Z_y} \|dG|_{TZ_y}\|_{HS}^2 d\|Z_y\|, \quad I = \int_Q E_y dy, \quad q = |Q|.$$

At a regular point of F , let $K = \ker dF$, and let $a \geq b$ be the singular values of $dF|_{K^\perp}$; then $ab = J_2F$. For any unit $w \in K$, the mixed two-vector $df(w) \wedge df(v)$, where v is a unit vector with $|dF(v)| = a$, has a component of norm $|dG(w)|a$. Since $\text{Dil}_2(f) \leq 1$, $|dG(w)|a \leq 1$, and hence $|dG(w)|^2 J_2F \leq b/a \leq 1$. Summing over an orthonormal basis of K gives

$$\|dG|_{\ker dF}\|_{HS}^2 J_2F \leq 2.$$

By coarea,

$$I \leq C \text{Vol}(R). \quad (7)$$

Let $\Omega = \{y \in Q : E_y \leq 4I/q\}$, so $|\Omega| \geq 3q/4$.

For real normal currents define

$$\text{EFill}_f(y) = \inf\{\text{Mass } C : C \in N_3(R, \partial R; \mathbb{R}), f_\#(\partial C) = \pm P_y\}.$$

This class is nonempty by filling the actual slice Z_y . If C is admissible, then $f_\#C$ fills P_y and $\text{Mass}(f_\#C) \leq \text{Mass } C$; Lemma 1 gives

$$\text{EFill}_f(y) \geq cS_1S_2S_3. \quad (8)$$

Therefore

$$A_\Omega := \int_\Omega \text{EFill}_f(y) dy \geq cS_1S_2S_3 q \geq cS_1S_2S_3^2S_4. \quad (9)$$

For $z \in Q$, let $\mathcal{B}_z$ be the set of $x \in R$ such that z belongs to the F -image of at least one coordinate 2-plaque through x . For fixed x , the set of such z 's has measure at most CR_3R_4 , because there are six plaques, each has area at most R_3R_4 , and $J_2F \leq 1$ on each plaque. Hence, for every normal 3-current B ,

$$\frac{1}{q} \int_Q \text{Mass}(B \lrcorner \mathcal{B}_z) dz \leq C \frac{R_3R_4}{S_3S_4} \text{Mass } B. \quad (10)$$

The following estimate is sufficient for the desired theorem. Take measurable near-minimizers C_y^0 with $\text{Mass } C_y^0 \leq 2 \text{EFill}_f(y)$; standard measurable selection in the separable flat topology, or equivalently outer integration with ε -minimizers, is enough for this formal step. One needs

$$A_\Omega \leq C \left(R_1 I + \frac{I^2}{S_1 q} \right) + \frac{C}{q} \int_Q \int_\Omega \text{Mass}(C_y^0 \lrcorner \mathcal{B}_z) dy dz. \quad (\text{EAT})$$

Assuming (EAT), (10) and $\text{Mass } C_y^0 \leq 2 \text{EFill}_f(y)$ give

$$A_\Omega \leq C \left(R_1 I + \frac{I^2}{S_1 q} \right) + C \frac{R_3R_4}{S_3S_4} A_\Omega.$$

If $R_3R_4 \leq \kappa S_3S_4$, the last term is absorbed by taking κ small. If also $R_1 \leq \kappa S_1$ and $I \leq \delta S_1 S_2^{1/2} S_3^{3/2} S_4$, then, since $q \sim S_3S_4$,

$$R_1 I \leq C \kappa \delta S_1 S_2 S_3^2 S_4, \quad \frac{I^2}{S_1 q} \leq C \delta^2 S_1 S_2 S_3^2 S_4.$$

For small δ, κ this contradicts (9). Thus $I \geq c S_1 S_2^{1/2} S_3^{3/2} S_4$, and (7) gives the desired second alternative.

Why the remaining estimate is non-routine

The estimate (EAT) cannot be obtained by a standard source-local cellular deformation in the quotient by f -null chains without a new idea. The obstruction is already visible in a two-cell model. Let A, A' be two oriented 2-cells with $f_\# A = f_\# A' = P$. Then $N = A - A' \in \ker f_\#$. If an exact-plaque bad set contains A but not A' , the source-local projection P_{bad} with $P_{\text{bad}} A = A$, $P_{\text{bad}} A' = 0$ gives $f_\# P_{\text{bad}} N = P \neq 0$. Dually, a quotient cochain has equal values on A and A' , but restricting it to the nonsaturated bad set destroys that property. Thus exact source localization and quotient invariance are in tension.

This does not disprove (EAT); it identifies what is missing. A proof would need an energy-sensitive, null-respecting carrier or a dual target-side cochain homotopy whose residual term is nevertheless controlled by the unsaturated exact-plaque mass in (10).

Remaining open issues

The document is therefore a conditional reduction, not a complete solution.

1. The essential mathematical gap is (EAT), an averaged tightening estimate for the degree-carrying part of the slices Z_y . It is used only in the last section, but it is indispensable for the high-aspect hard case.
2. The first alternative, the single-slice bound, the weighted coarea estimate, the essential filling lower bound, the exact-plaque Fubini estimate, and the absorption algebra are proved above.
3. Standard approaches tried so far do not close the gap: monomial degree estimates give only (1); ordinary source deformations create uncontrolled f -null boundary errors; quotient cochains are not compatible with nonsaturated exact-plaque localization as shown above.
4. A complete proof would require either a proof of (EAT), an equivalent dual formulation using only target-pullback cochains, or a different hard-regime argument exploiting the full two-parameter family of slices and the ambient mixed 2-Jacobian bounds.

References

- [1] L. Guth, *Area-expanding embeddings of rectangles*, arXiv:0710.0403.
- [2] L. Guth, *Area-contracting maps between rectangles*, Ph.D. thesis, Massachusetts Institute of Technology, 2005.
- [3] L. Guth, *Directional isoperimetric inequalities and rational homotopy invariants*, arXiv:0802.3549.